

Let N be a compact connected oriented hypersurface in $\mathbb{C}H^2$ which is quadratically convex, that is, the shape operator at each point is definite. In particular, by [2], it is the border of a convex body and diffeomorphic to a sphere. We call U the exterior of N , that is, the connected component of $\mathbb{C}H^2 - N$ into which the opposite of the mean curvature field points.

We call J the complex structure of $\mathbb{C}H^2$. Given a unit tangent vector u we denote by γ_u the unique geodesic of $\mathbb{C}H^2$ whose initial velocity is u . Let $F : N \times \mathbb{R} \rightarrow \mathbb{C}H^2$ be the smooth map defined by

$$F(q, r) = \gamma_{J\nu(q)}(r),$$

where ν is the unit normal vector field pointing inwards, that is, in the direction of the mean curvature vector field of N . This is our first result:

Theorem 1 *Let N be a quadratically convex compact connected oriented hypersurface of $\mathbb{C}H^2$. Then, the restrictions of F to $N \times (0, \infty)$ and $N \times (-\infty, 0)$ are diffeomorphisms onto the exterior U of N .*

Thus, we have a **double geodesic ray foliation** of U and this allows us to define well an **outer billiard map**, as follows: For $t > 0$,

$$B : U \rightarrow U, \quad B(\gamma_{J\nu(p)}(-t)) = \gamma_{J\nu(p)}(t).$$

As a direct consequence of Theorem 1, we have:

Theorem 2 *The outer billiard map B is a diffeomorphism.*

Remark 3 *In general, for a Hadamard manifold (a simply connected manifold with nonpositive sectional curvature), weaker hypotheses on the convexity of the hypersurface may allow the outer billiard map to be well defined without being smooth. For instance, Proposition 1.3 of [9] presents an example of a strictly convex surface S in $\mathbb{R}^3$ (that is, for each $p \in S$, the affine plane tangent to S at p intersects S only at p , near p), for which the outer billiard map does exist, but it is not smooth.*

The following proposition provides examples. Part (b) is a particular case of part (a). We recall that a complex line in $\mathbb{C}H^2$ is a complete totally geodesic surface which is a complex submanifold (its tangent spaces are invariant by J). It is isometric to the real hyperbolic plane $H^2(-4)$ of constant curvature -4 . A complex line admits an isometric reflection in $\mathbb{C}H^2$ with respect to it.